

Koushik Rao Gourineni

Data Engineer

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PROFESSIONAL SUMMARY

Data Engineer with 5+ years of experience building scalable **data pipelines** and **real-time streaming systems** processing **400K+ events per minute**. Expertise in **ETL/ELT pipeline development** using **PySpark**, **Apache Spark** and **Apache Kafka** on **AWS** and **GCP**, reducing **analytics latency** from hours to minutes and enabling **data-driven decision making** for **200+ stakeholders**. Skilled in designing **data warehouses** using **Snowflake** and **Amazon Redshift**, implementing **Lambda architecture** and deploying **containerized data pipelines** using **Docker** and **Kubernetes**. Experienced in **Apache Airflow orchestration**, **data modeling**, **data quality monitoring** and building **scalable analytics platforms** supporting **fraud detection**, **customer analytics** and **business intelligence**.

TECHNICAL SKILLS

Programming: Python (PySpark, pandas, confluent-kafka), SQL, Shell Scripting

Big Data & Processing: Apache Spark (Structured Streaming, Batch Processing), Apache Kafka, dbt, Delta Lake

Data Engineering: ETL/ELT Pipeline Development, Data Modeling (Dimensional, Star Schema), Data Lake Architecture (Medallion, Lambda), Schema Design, Data Quality Frameworks (Great Expectations), Feature Engineering

Workflow Orchestration: Apache Airflow, Airflow DAG Development

Cloud Platforms: AWS (EMR, S3, Redshift, EKS, Glue, CloudWatch), GCP (GKE, Cloud Composer, Cloud Storage, Cloud Bigtable, Cloud Monitoring)

Databases & Data Warehouses: Snowflake, Amazon Redshift, PostgreSQL, MySQL

DevOps & Infrastructure: Docker, Kubernetes (EKS, GKE), Helm, Jenkins CI/CD, Git, GitHub

Data Visualization: Tableau, Power BI, Looker, SQL-based Analytics

Methodologies: Agile/Scrum, Stream Processing, Batch Processing, Performance Tuning, Data Quality Testing, Statistical Data Validation

EXPERIENCE

PayPal – Data Engineer

Location: California (Remote), USA

Duration: April 2024 – Present

- Built **real-time data pipelines** using **PySpark Structured Streaming**, **Apache Kafka** and **Google Kubernetes Engine (GKE)** processing **170K+ payment events/min (400K peak)**, reducing **analytics latency** from **6 hours to under 5 minutes**.
- Developed **stream processing jobs** to enrich **Kafka transaction streams** with **merchant metadata** and **currency conversion data**, implementing **30-second micro-batches** and **event-time watermarking** to handle **late-arriving events**.
- Optimized **PySpark streaming** performance by replacing **shuffle-heavy joins** with **broadcast joins**, reducing **micro-batch processing time** from **48s to 19s** and sustaining **400K events/min throughput**.
- Implemented **stateful stream processing** using **PySpark mapGroupsWithState** to generate **60+ ML features** across **1-hour, 24-hour, 7-day, and 30-day windows** for **real-time fraud detection models**.
- Built a **data warehouse** using **dbt** and **Snowflake** processing **14B+ transaction records**, creating **staging, intermediate** and **data mart layers** to support **200+ analysts and product teams**.
- Deployed **containerized data pipelines** using **Docker**, **Helm** and **GKE**, orchestrated workflows with **Apache Airflow (Cloud Composer)** and implemented **data quality validation** with **Great Expectations** and **PagerDuty monitoring**.

Freshworks – Data Engineer

Location: India

Duration: March 2019 – December 2022

- Designed **Lambda architecture data pipelines** processing **50M+ daily events** using **Apache Kafka**, **PySpark** and **Amazon EMR**, reducing **BI dashboard refresh time** from **20+ hours to under 15 minutes** across a **Redshift data warehouse** supporting **40M+ customer accounts**.
- Built **batch and streaming ETL pipelines** using **PySpark** to process **120M+ daily product events**, implementing **sessionization logic** and storing optimized **Parquet datasets in Amazon S3**, enabling **product analytics in Snowflake** for **8M monthly active users**.
- Developed **Kafka consumer microservices in Python (confluent-kafka)** for **near real-time ingestion**, writing **5–10 minute micro-batches to Amazon S3**, deployed via **Docker on Amazon EKS** with **Prometheus monitoring** and **auto-scaling**.
- Implemented **real-time analytics pipelines** using **PySpark Structured Streaming**, performing **stateful aggregations and joins** to compute **customer health scores** with **under 10-minute latency** for **12K enterprise accounts**.
- Built and optimized **50+ scalable ETL workflows** with **PySpark**, applying **data validation**, **schema enforcement**, **deduplication using window functions**, and loading curated datasets into **Amazon Redshift** and **Snowflake**.
- Orchestrated **end-to-end data workflows** using **Apache Airflow DAGs**, integrating **EMR jobs**, **data quality checks** and **Snowflake loads**, delivering datasets for **SLA reporting**, **product analytics** and **churn modeling**.

EDUCATION

Master's in Information Technology and Management, Florida Atlantic University, FL, USA.